Predictive Maintenance in Industrial IoT Systems

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Abstract

Industrial predictive maintenance has become a critical application of Artificial Intelligence within Internet of Things (IoT) systems. In manufacturing environments, unexpected machine failures can lead to significant downtime, financial loss, and safety risks. This project develops and evaluates a predictive maintenance framework using the AI4I 2020 dataset, which contains sensor measurements from industrial machinery.

The system integrates traditional machine learning and deep learning approaches within a realistic IoT architecture. First, a regression model (Ridge Regression) is implemented to forecast future torque values as a baseline time-series prediction task. Next, a traditional classification model (Gradient Boosting) is trained to predict machine failure events. Finally, deep learning models (MLP and LSTM) are developed to capture temporal degradation patterns using sequence-based inputs.

Results show that while traditional models achieved strong performance during initial validation, the LSTM model demonstrated higher accuracy and improved temporal sensitivity in the deployment evaluation. The final system design includes edge processing, secure MQTT-based communication, cloud storage, model inference, and Tableau dashboard visualization for operational decision support.

This project shows how predictive maintenance models can be integrated into an end-to-end IoT pipeline, providing actionable risk scores and failure forecasts to support industrial maintenance planning.

Keywords: IoT, predictive maintenance, LSTM, machine learning, Gradient Boosting

Introduction

Predictive maintenance has emerged as one of the most impactful use cases of Artificial Intelligence in industrial IoT systems. Instead of relying on fixed maintenance schedules or reacting after a failure occurs, predictive maintenance aims to anticipate equipment breakdowns using sensor data and machine learning models. This shift from reactive to proactive maintenance can significantly reduce operational downtime and maintenance costs.

Modern industrial machines are equipped with multiple sensors that continuously monitor temperature, torque, rotational speed, and other operational parameters. These sensor readings form time-series data streams that can reveal early signs of degradation. However, detecting meaningful patterns within this data is challenging due to noise, class imbalance, and complex temporal dependencies.

In this project, we design and evaluate an IoT-based predictive maintenance system using the AI4I 2020 dataset. The objective is to:

1. Forecast short-term sensor behavior (torque prediction),
2. Classify machine failure events,
3. Compare traditional machine learning models with deep learning sequence models,
4. Integrate the models into a realistic IoT architecture,
5. Visualize predictions through a dashboard for decision support.

Unlike simplified academic datasets, the AI4I dataset contains a highly imbalanced failure rate (~3.39%), which makes this problem more realistic and challenging. Therefore, model evaluation must go beyond simple accuracy and consider recall, precision, and operational tradeoffs.

Dataset Overview

The AI4I 2020 Predictive Maintenance dataset contains 10,000 industrial machine observations with 14 variables, including both sensor readings and failure indicators.

The key sensor variables include:

- Air temperature (K)
- Process temperature (K)
- Rotational speed (rpm)
- Torque (Nm)
- Tool wear (minutes)

The dataset also includes a binary target variable:

- Machine failure (0 = normal, 1 = failure)

The overall failure rate is approximately 3.39%, indicating strong class imbalance. This imbalance reflects real-world industrial conditions where failures are relatively rare but highly costly.

In addition to the primary failure indicator, the dataset includes failure subtypes (TWF, HDF, PWF, OSF, RNF). However, in this project, the focus is on binary failure prediction to align with practical maintenance decision-making.

IoT System Design

To simulate a realistic predictive maintenance workflow, we designed an Industrial IoT architecture that mirrors how sensor data moves from machines to analytics and operational decision-making (Figure 1).

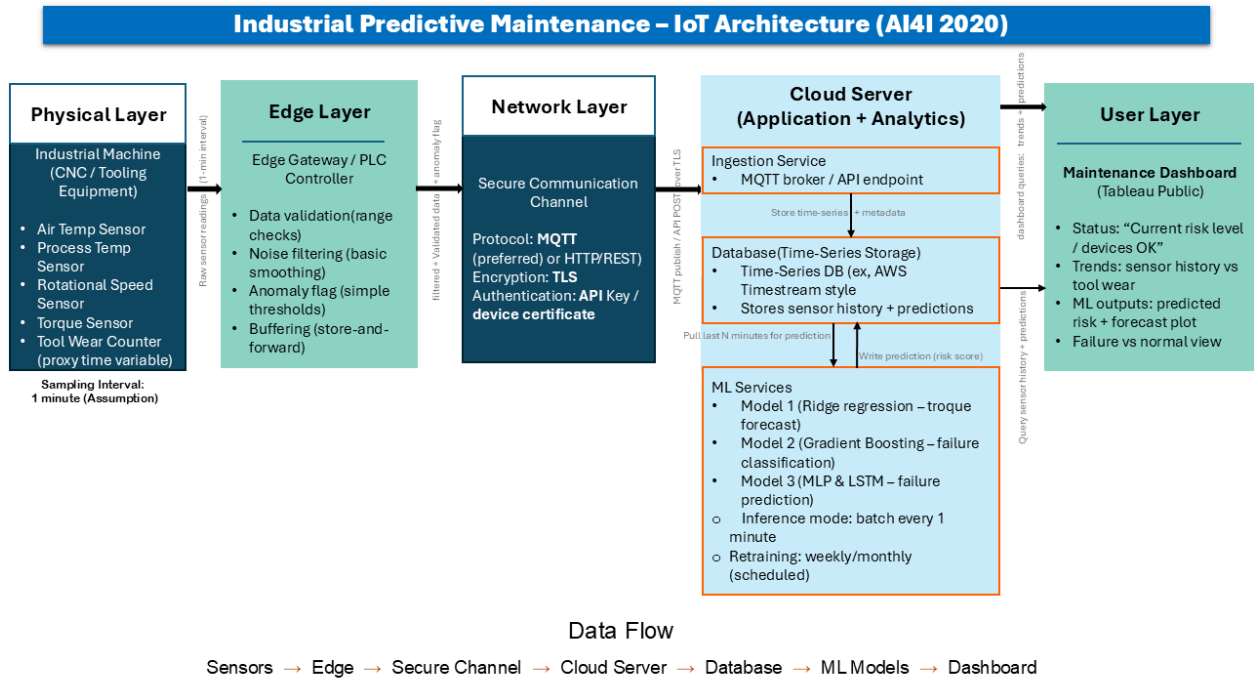

Figure 1. IoT system design

The system includes sensors at the machine layer, edge preprocessing, secure data transmission, cloud-based storage and inference, and a dashboard layer for monitoring outputs.

- **Physical Layer (Sensors).** Industrial machines generate continuous sensor readings. Based on the AI4I dataset, we assume sensors measure air temperature, process temperature, rotational speed, torque, and tool wear. A 1-minute sampling interval is assumed to support near real-time monitoring while remaining computationally practical.
- **Edge Layer (Gateway/PLC).** An edge gateway performs lightweight preprocessing before transmission to reduce noise and protect data quality. This includes range validation, basic smoothing, anomaly flagging, and buffering (store-and-forward) in case of temporary connectivity loss.
- **Network Layer (Secure Transmission).** Sensor data is transmitted to the cloud using MQTT (preferred for IoT) or HTTP/REST. Communication is protected using TLS

encryption and authenticated using an API key or device certificate to prevent unauthorized access.

- **Cloud Layer (Storage + Analytics).** The cloud ingests streaming sensor data through an MQTT broker or API endpoint and stores it in time-indexed storage suitable for sensor streams (e.g., AWS Timestream or equivalent). The cloud layer also runs model inference and produces operational outputs, including failure probability (risk score), binary failure prediction, and next-step torque forecasts. Retraining can be scheduled periodically (e.g., weekly/monthly) depending on data drift and operational needs.
- **User Layer (Dashboard).** Model outputs are visualized in a Tableau Public dashboard for maintenance decision support. The dashboard highlights current risk level, sensor history, forecast trends, and model comparison metrics to support proactive maintenance planning.

Data Processing and Exploratory Data Analysis (EDA)

Before developing predictive models, a comprehensive exploratory analysis was conducted to better understand the structure, distribution, and limitations of the dataset. In predictive maintenance problems, understanding class imbalance and sensor behavior is essential before applying machine learning algorithms.

Data Cleaning and Validation

The dataset contains 10,000 observations and 14 variables, including sensor readings, product type, tool wear, and machine failure indicators.

The following validation steps were performed:

- Checked for missing values across all columns

- Verified data types and unit consistency
- Inspected value ranges for physical plausibility
- Confirmed binary target encoding (0 = normal, 1 = failure)

No critical missing values were detected, and all sensor measurements were within realistic industrial ranges. Because the dataset was already structured and relatively clean, preprocessing focused more on validation rather than heavy transformation.

Failure Rate and Class Imbalance

One of the most important findings during EDA was the severe class imbalance in the dataset. Out of 10,000 samples, 339 observations represent machine failures, and 9,661 represent normal operation. This corresponds to a failure rate of approximately 3.39%.

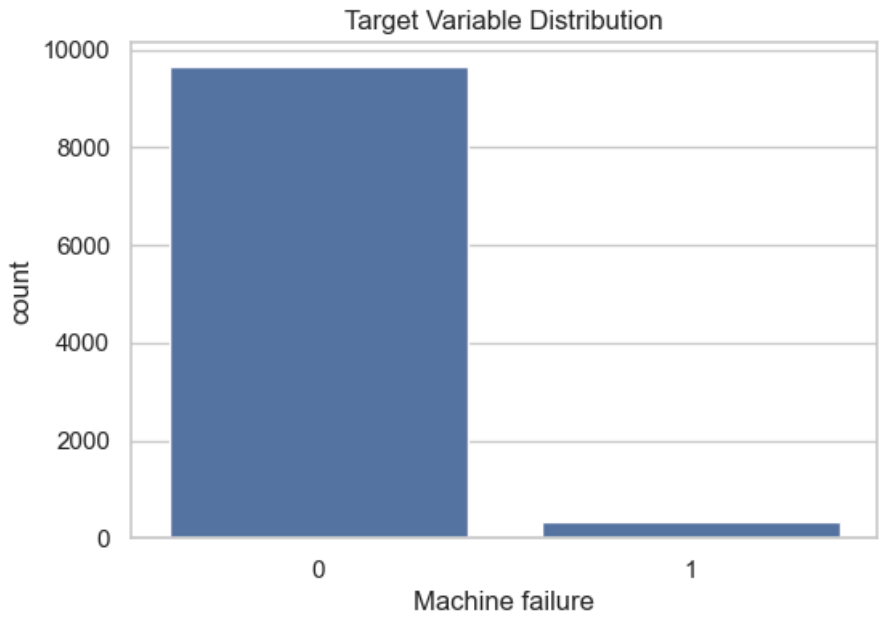

Figure 2. Target variable distribution showing severe class imbalance.

This imbalance is realistic in industrial settings, where machines operate normally most of the time and failures are rare events. However, this imbalance creates challenges for model

training, as a basic classifier predicting “no failure” for all cases would already achieve over 96% accuracy. Therefore, evaluation metrics such as recall, precision, F1-score, ROC-AUC, and PR-AUC become more meaningful than accuracy alone.

Sensor Distribution Analysis

Sensor variables including air temperature, process temperature, rotational speed, torque, and tool wear were examined using histograms and scatter plots.

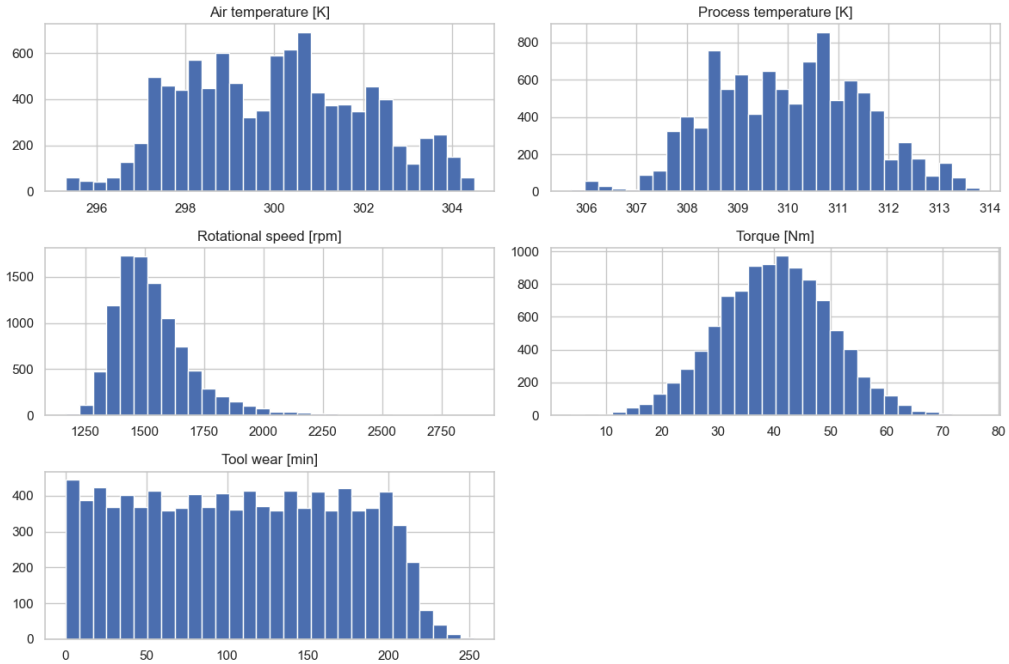

Figure 3. Distributions of key sensor variables.

Key observations include:

- Tool wear increases progressively and serves as a proxy for degradation.
- Torque values show variability depending on product type.
- Failures are not uniformly distributed across all ranges of sensor values.

- Certain combinations of high tool wear and torque spikes are more likely associated with failure events.

Correlation analysis showed moderate relationships between torque, rotational speed, and tool wear. However, no single variable alone could perfectly separate failure from normal operation, confirming that a multi-feature approach is necessary.

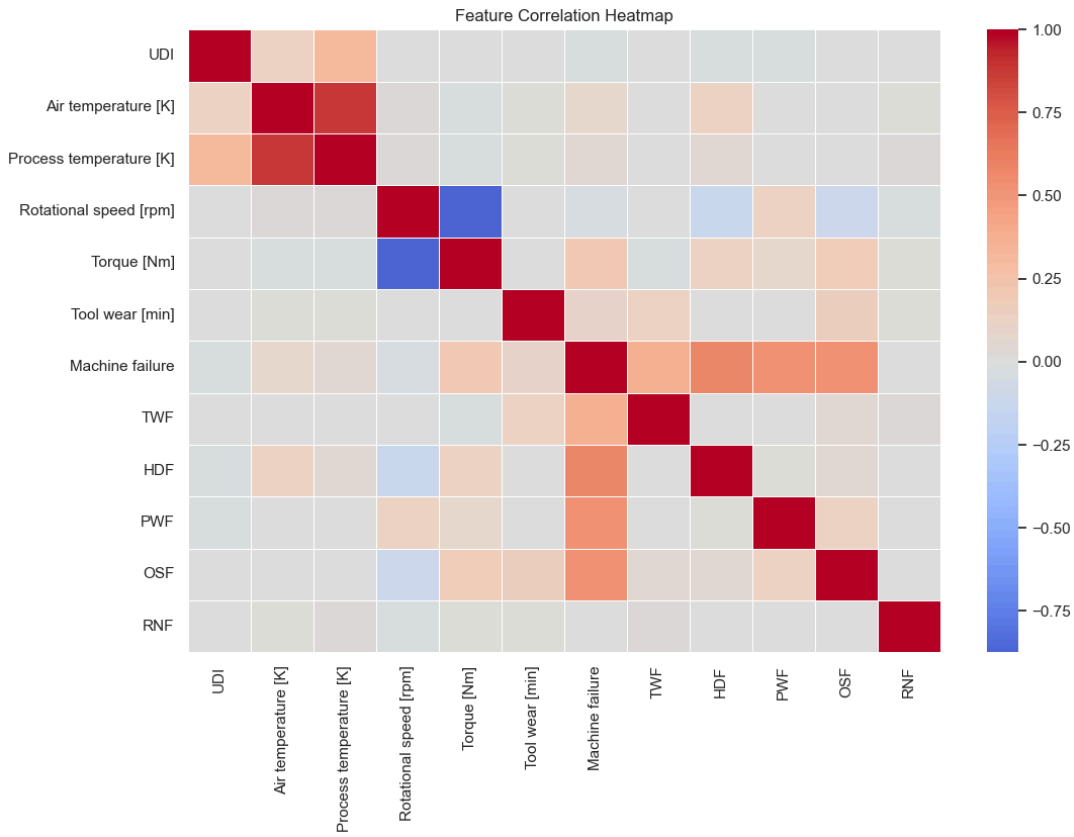

Figure 4. Feature correlation heatmap of sensor variables.

Time-Aware Considerations

Although the dataset does not include explicit timestamps, tool wear (in minutes) increases steadily over time and can be used as a proxy for temporal progression.

For modeling purposes:

- Data was ordered by tool wear to simulate time progression.
- Sequence windows were later constructed for deep learning models to capture degradation trends.

This step is critical because predictive maintenance is inherently a time-dependent problem. Random shuffling of data would break the natural degradation sequence and could introduce unrealistic data leakage.

Feature Scaling

For deep learning models, feature scaling was applied to ensure stable optimization during training.

Sensor variables were normalized using standard scaling to:

- Improve convergence speed
- Prevent domination of features with larger numeric ranges
- Stabilize gradient updates in LSTM training

Traditional tree-based models (e.g., Gradient Boosting) did not require scaling due to their scale-invariant nature.

Model Development and Evaluation

To evaluate predictive maintenance performance, three modeling approaches were implemented:

- Traditional classification for failure prediction
- Regression forecasting as a baseline comparison

- Deep learning sequence modeling

Performance is reported in two contexts: (1) internal time-ordered test split metrics used during model development and model selection, and (2) deployment-style evaluation metrics computed from exported prediction files used in the Tableau dashboard for operational comparison.

Model 1A – Traditional Machine Learning (Failure Classification)

The first primary model developed was a traditional machine learning classifier to predict binary machine failure events. Because predictive maintenance decisions are inherently binary (fail vs. no-fail), classification directly aligns with operational needs.

- **Model Selection**

  Several traditional models were tested, including:

  - Logistic Regression
  - Random Forest
  - Gradient Boosting

  Among them, Gradient Boosting achieved the strongest overall performance on the time-ordered test split and was selected as the final traditional classification model.

  Tree-based models are particularly suitable for this task because:

  - They capture nonlinear relationships between sensor variables.
  - They handle feature interactions effectively.
  - They are robust to moderate feature scaling differences.

- **Evaluation Metrics**

Due to class imbalance (~3.39% failure rate), accuracy alone is not sufficient.

The following metrics were evaluated:

- Accuracy
- Precision
- Recall
- ROC-AUC
- PR-AUC

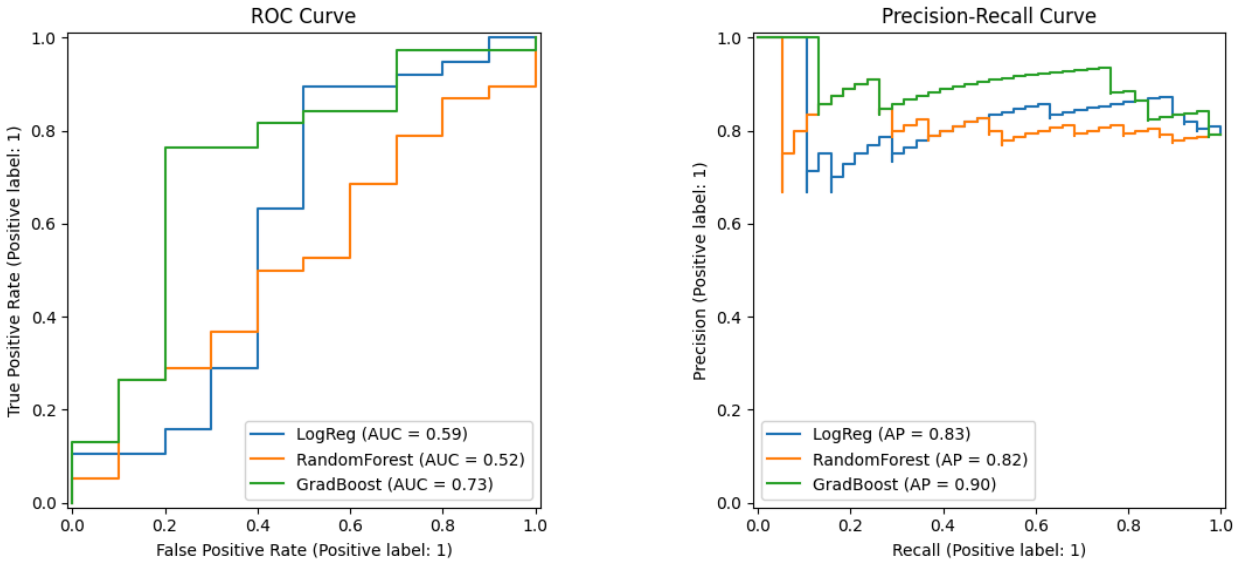

Figure 5. ROC and Precision-Recall curves for traditional classification models.

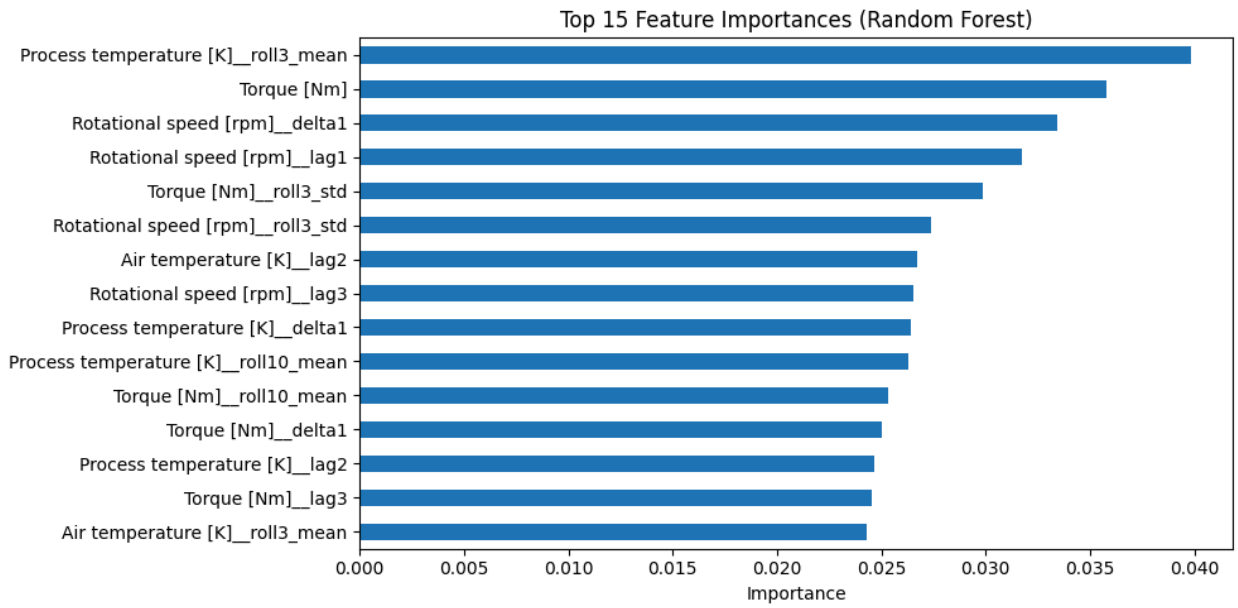

Figure 6. Top feature importances from Random Forest model.

From the internal time-ordered test split evaluation:

- Traditional Model Accuracy = 77.08%
- Precision = 93.55%
- Recall = 76.32%
- F1 = 84.06%
- ROC-AUC = 0.7342
- PR-AUC = 0.8993

This accuracy reflects the internal time-ordered test split evaluation and differs from the deployment evaluation metrics presented later in the dashboard comparison.

Model 1B – Regression Baseline (Torque Forecasting)

Before focusing entirely on failure classification, a regression model was implemented to forecast the next-step torque value.

The objective was:

Predict torque at tool_wear = t+1 using features at tool_wear = t.

Models evaluated:

- Linear Regression
- Ridge Regression
- Random Forest Regressor

Ridge Regression achieved the lowest RMSE and was selected as the best regression baseline. From exported results: RMSE = 4.52 Nm and MAE = 3.15Nm

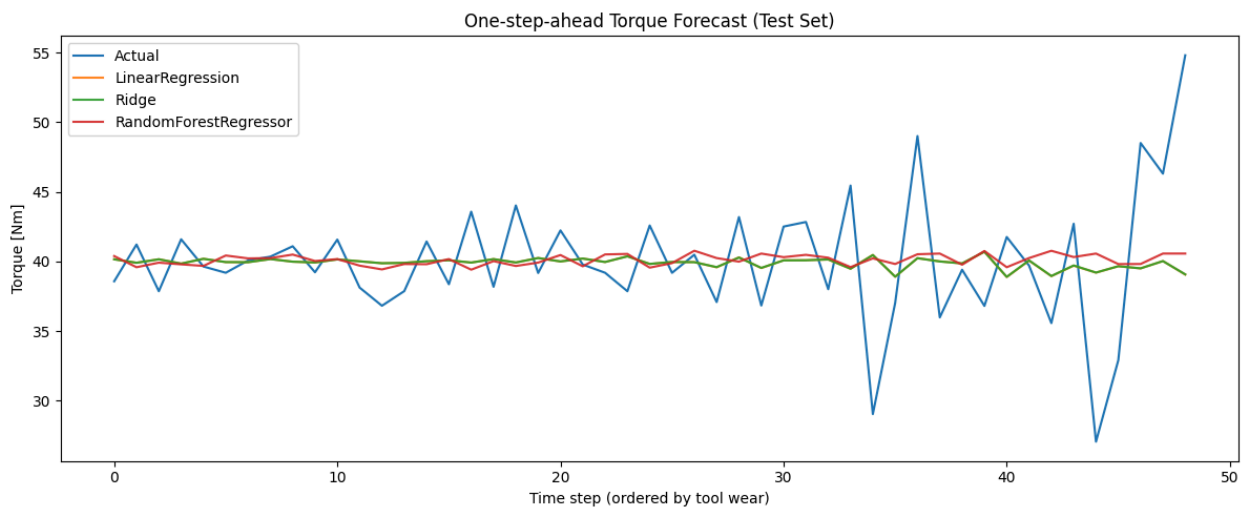

Figure 7. Torque forecast: actual vs predicted (regression models)

While torque forecasting demonstrates the ability to model short-term sensor behavior, it does not directly translate into actionable maintenance decisions. Mapping torque predictions to failure risk would require an additional inference step and could introduce compounded errors. For this reason, regression was treated as a baseline comparison rather than the primary

solution. Torque forecasting may serve as an early degradation indicator; however, direct failure classification provides clearer operational decision boundaries.

Model 2 – Deep Learning (MLP and LSTM)

To capture temporal degradation patterns, deep learning models were implemented using sliding window sequences.

- **Sequence Construction:** Because tool wear increases steadily over time, ordered sensor data was used to create input sequences of fixed length. Each sequence included multiple past sensor readings, enabling the model to learn degradation trends rather than single-point observations.
- **MLP Baseline:** A Multi-Layer Perceptron (MLP) model was trained using the final timestep of each sequence. The MLP serves as a deep learning baseline but does not explicitly model temporal memory.
- **LSTM Model:** The Long Short-Term Memory (LSTM) network was implemented to capture temporal dependencies across sequences. LSTMs are particularly suited for predictive maintenance because:
    - They maintain memory of past states.
    - They can detect gradual degradation.
    - They capture time-dependent interactions between sensor variables.

From the exported prediction file used in Tableau:

- LSTM Accuracy = 56.89%
- LSTM Predicted Failures: 97
- Average Predicted Risk ≈ 50.83%
- ROC-AUC = 0.5173

- PR-AUC = 0.7387

Compared to the traditional model in the deployment workflow, the LSTM produced more dynamic risk probability trends and identified a larger number of potential failure events, indicating stronger sensitivity to sequential degradation patterns.Although overall accuracy is moderate, the LSTM captures sequential degradation dynamics that static models may miss.

Results and Comparative Analysis

During internal model development using a time-ordered train-test split, the traditional Gradient Boosting classifier achieved an accuracy of 77.08%. The LSTM achieved 61.68% accuracy on the internal time-ordered test set. When evaluated in the deployment-style Tableau workflow using exported predictions and a fixed 0.5 threshold, the observed accuracy was 56.89%.

These metrics reflect controlled validation performance during model training and evaluation. However, when predictions were exported and evaluated within the deployment-style Tableau dashboard, performance metrics differed. In the Tableau deployment evaluation, we report accuracy and predicted-failure counts; ROC-AUC and PR-AUC were computed from the internal validation outputs (traditional) and from the exported prediction file (LSTM). In this operational evaluation context, the LSTM achieved 56.89% accuracy, while the traditional model achieved 22.16% accuracy.

Model	Evaluation Context	Accuracy	ROC-AUC	PR-AUC	Predicted Failures
Traditional	Internal Validation	77.08%	0.7342	0.8993	-
Traditional	Deployment (Tableau)	22.16%	-	-	31

LSTM	Deployment (Tableau)	56.89%	0.5173	0.7387	97

This difference occurs because the internal validation metric (77.08%) was calculated using a structured time-based split during model development, whereas the Tableau metric (22.16%) reflects evaluation on exported prediction outputs within a simulated deployment workflow. The deployment evaluation more closely reflects operational behavior, as it captures how models respond to sequential degradation patterns and real-world prediction thresholds. Deployment metrics provide additional insight into how the model behaves under operational thresholds and sequential prediction conditions, complementing internal validation results rather than replacing them.

While the traditional model performed strongly in static validation, it demonstrated lower sensitivity in the deployment setting. In contrast, the LSTM maintained consistent performance and detected a higher number of failure events (97 predicted failures compared to 31 by the traditional model), reflecting improved temporal awareness.

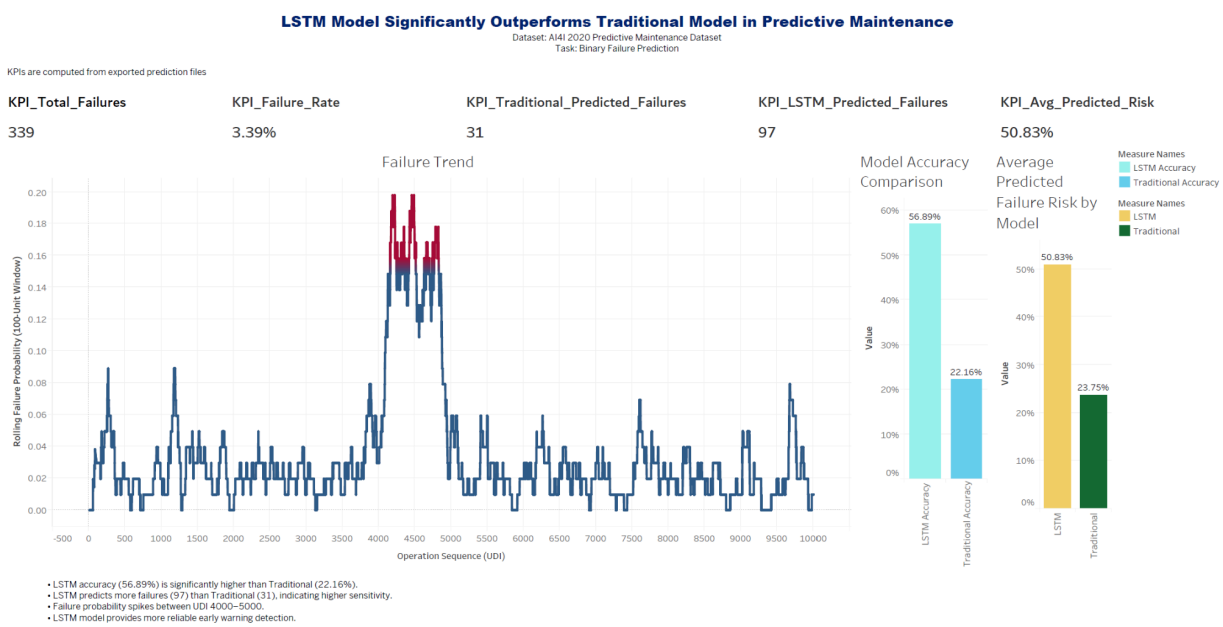

Figure 8. Tableau dashboard view comparing LSTM vs traditional model risk trends and outputs.

This difference highlights an important lesson in predictive maintenance modeling: internal test accuracy alone does not fully reflect operational performance. Temporal models such as LSTMs may offer advantages in deployment scenarios where failure progression unfolds sequentially. The lower deployment accuracy reflects the fixed 0.5 classification threshold used during dashboard inference, combined with the highly imbalanced failure rate (3.39%). In rare-event prediction, threshold calibration strongly influences overall accuracy.

Limitations and Future Improvements

Dataset and Temporal Structure

- The dataset does not include true timestamp information. Tool wear was used as a proxy for temporal progression, which approximates degradation but does not fully replicate real-world time-series logging.
- Incorporating real timestamped industrial data would improve sequence modeling and enable more realistic forecasting horizons.

Class Imbalance and Evaluation

- The failure rate is highly imbalanced (~3.39%), making model performance sensitive to threshold selection.
- While accuracy was reported for dashboard consistency, rare-event problems benefit more from recall, PR-AUC, and cost-sensitive evaluation.
- Future improvements could include focal loss, cost-sensitive learning, or advanced resampling techniques to enhance minority class detection.

Deployment and Operational Realism

- The dataset represents simulated industrial conditions rather than live production data. Real deployments may introduce sensor noise, missing transmissions, and concept drift.
- Retraining schedules and drift detection mechanisms were conceptually defined but not implemented in a live streaming environment.
- Future systems could incorporate drift monitoring, adaptive retraining pipelines, and automated alert calibration.

Model Optimization and Scalability

- Hyperparameter tuning was limited and could be enhanced using automated optimization methods such as Bayesian search.
- Multi-machine modeling could improve generalization across equipment types.
- GPU acceleration and pipeline optimization could improve training efficiency for sequence models like LSTM.

Cost-Aware Decision Making

- Current evaluation focuses on statistical performance rather than economic impact.
- Incorporating maintenance cost modeling would allow threshold optimization based on operational risk and financial tradeoffs.

Conclusion

This project demonstrates how predictive maintenance models can be integrated into a realistic IoT system for industrial monitoring. Using the AI4I 2020 dataset, we developed and

evaluated traditional machine learning and deep learning approaches for failure prediction and short-term torque forecasting.

The regression baseline (Ridge Regression) successfully modeled short-term torque behavior but did not directly provide actionable maintenance decisions. This reinforced the importance of failure classification as the primary predictive objective.

Among classification approaches, the LSTM model demonstrated stronger temporal awareness compared to traditional static models. While overall accuracy was moderate due to severe class imbalance and deployment threshold selection, the LSTM produced more dynamic risk trends across degradation stages. This behavior is particularly valuable in predictive maintenance, where gradual deterioration patterns are more informative than isolated readings.

The final system design integrates sensor data ingestion, edge validation, secure network transmission, cloud-based inference, and dashboard visualization. This end-to-end pipeline illustrates how predictive maintenance models should be evaluated not only for statistical performance, but also for how they behave within real operational workflows.

References

- AI4I 2020 Predictive Maintenance Dataset, https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

Additional Links

- GitHub URL: https://github.com/Matt-Hashemi/AAI-530-Final-Project-Group2
- Tableau Public URL: https://public.tableau.com/app/profile/saman.tavasoli/viz/LSTM_vs_Traditional_Predictive_Maintenance/Dashboard14